ETFs Simulation with Temporal Memory: A Positive Fractional–Diffusive Volatility Model

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Abstract

We develop a conditional-variance model that weights past squared shocks with a finite fractional kernel and pools them across assets through a positive correlation-graph resolvent. We derive its positivity and fixed-parameter moment properties, exact multi-step variance forecasts, and the effective memory of a persistent asymmetric extension. The empirical study covers the S&P 500, eleven sector ETFs and eight country ETFs, with 186 origins per panel during 2023–2026 and horizons of 1, 5 and 21 observations. Persistence, asymmetry and multi-horizon selection improve the initial specification in all six US panel–horizon comparisons. A fixed 50:50 combination with GJR-GARCH lowers pooled QLIKE relative to the refined fractional model and FIGARCH in all nine comparisons. Against GJR alone, it has lower loss in three; all nine paired nominal 95% block-bootstrap intervals include zero at both block lengths. The study therefore establishes neither superiority nor equivalence to GJR. Diffusion has mixed value, and the ranking against weighted Network GARCH reverses between sectors and countries. Evaluation uses retrospective adjusted-price snapshots downloaded in October 2026; chronological row access does not reconstruct historical data availability. The shared temporal kernel and graph operator make the model's impulse response tractable, while the ablations show where these restrictions help or hinder forecasting.

Keywords: fractional memory; conditional variance; graph diffusion; sector ETFs; volatility forecasting; chronological validation.

1 Introduction

Volatility forecasts can draw on an asset's past shocks and on shocks to related assets. We combine fractional temporal weighting and correlation-graph diffusion in a conditional-variance model, with separate parameters for memory, cross-asset smoothing and feedback.

Fractional differencing [9] and FIGARCH [1] motivate the use of slowly decaying temporal weights. Our construction uses a finite, normalized fractional-integral kernel on squared standardized innovations. An implicit diffusion step smooths these signals over the graph [19]; a positive intercept and subunit feedback complete the recursion.

Other parsimonious volatility models distribute information across horizons. The HAR-RV model of Corsi [5] combines daily, weekly and monthly realized-volatility components. Bollerslev et al. [4] show how accounting for variation in measurement error can improve

realized-volatility forecasts. Our daily-close data preclude fitting these high-frequency benchmarks, but their results motivate attention to both memory structure and proxy quality.

1.1 Position relative to network volatility models

Spatial ARCH permits conditional heteroskedasticity to depend on neighboring locations [14], and network log-ARCH combines temporal effects with contemporaneous network interactions [11]. Network GARCH uses graph structure to reduce the dimension of multivariate volatility estimation, with stationarity and QMLE results [22]. Threshold network GARCH also incorporates asymmetric shock responses [15]. More recently, generalized network GARCH includes neighboring shock and variance terms alongside covariance forecasts [23]. The contribution here is the particular restriction to a common positive resolvent at every lag and a normalized fractional temporal profile. This defines a restricted multivariate ARCH recursion distinct from FIGARCH. We fit FIGARCH and first-order Network GARCH benchmarks (Sections 8–8.4); network log-ARCH, threshold Network GARCH and DCC-GNGARCH remain outside the comparison. The proofs apply positive-filter and graph arguments to this parameterization, whose network dependence and asymmetric responses are established ingredients.

Contributions. We specify a positive fractional–diffusive variance family, establish its fixed-parameter moment properties and derive its effective memory and exact multi-step forecasts. We assess its restrictions through ablations, conventional and network benchmarks, forecast combination and calibration diagnostics. A fixed-graph bias–variance calculation identifies when diffusion reduces signal-estimation error (Appendix F); a separate forward protocol tests the mechanism on future range-based targets (Appendix G). Appendix A describes reproduction using the retained price snapshots.

Roadmap. Sections 2–6 define and evaluate the initial specification; Section 7 adds persistence and asymmetry; Section 8 evaluates country ETFs and additional benchmarks; Section 9 interprets the findings, and the appendices give reproduction details, asset paths and additional diagnostics.

2 Data, notation and prediction target

2.1 Observed prices and returns

Let $P_{t,j} > 0$ denote the closing price of asset j at retained trading observation t, and define

$$r_{t,j} = \log(P_{t,j}/P_{t-1,j}). \tag{1}$$

Write $\mathcal{F}_o = \sigma(P_0, \ldots, P_o)$ for the information generated by observations through origin o in a fixed data vintage. All transformations, graph estimates and parameter choices used to forecast from o depend only on these rows. For the empirical study, this ordering

refers to the downloaded snapshot; it does not establish which versions of those rows were available on date o.

The market snapshot was downloaded on 17 October 2026 from Yahoo Finance through `yfinance` 0.2.66, with adjusted prices and an exclusive request end date of 17 October 2026. It contains the index `^GSPC` and the eleven ETFs XLB, XLE, XLF, XLI, XLK, XLP, XLRE, XLU, XLV, XLY and XLC. After inception trimming and alignment, common prices run from 19 June 2018 through 16 October 2026. Prices must be positive, finite, uniquely dated and increasing in date; missing interior observations are not filled.

The index and ETF panel are fitted separately. Adjusted ETF returns and an index price-return series represent different economic quantities. The index is therefore a separate forecasting series, rather than a self-financing replication by the sector funds. Section 9.2 explains how price revisions can affect this retrospective evaluation.

2.2 Risk target and common mean convention

At each parameter refit, estimate a mean μ_j and a positive scale s_j^2 from the observed fitting window. These estimates are shared by all benchmark models until the next refit. Write

$$u_{t,j} = r_{t,j} - \mu_j, \qquad z_{t,j} = u_{t,j}^2 / s_j^2. \tag{2}$$

The h-observation evaluation target is

$$Y_{o,h,j} = \sum_{\ell=1}^{h} (r_{o+\ell,j} - \mu_j)^2, \qquad h \in \{1, 5, 21\}. \tag{3}$$

The target uses daily returns. Under the specified constant conditional mean, it is conditionally unbiased for the sum of future conditional variances. A misspecified mean adds squared mean error to that expectation. The empirical task therefore measures forecasts of centered return second moments. Intraday realized variance aggregates high-frequency returns and has its own measurement-error theory [2]; our daily squared-return proxy does not provide that richer measurement.

3 A positive fractional–diffusive model

3.1 Finite fractional-integral memory

For $0 < d \leq 1$, the binomial expansion of $(1 - B)^{-d}$ yields positive coefficients

$$g_0(d) = 1, \qquad g_k(d) = \frac{k - 1 + d}{k}\, g_{k-1}(d) = \frac{\Gamma(k + d)}{\Gamma(d)\Gamma(k + 1)}. \tag{4}$$

With memory length M, normalize these coefficients:

$$p_k(d) = \frac{g_k(d)}{\sum_{v=0}^{M-1} g_v(d)}, \quad m_t = \sum_{k=0}^{M-1} p_k(d) z_{t-1-k}. \tag{5}$$

The weights satisfy $p_k > 0$ and $\sum_k p_k = 1$. For large k, $g_k(d)$ behaves as $k^{d-1}/\Gamma(d)$. Within a fixed window, increasing d spreads weight toward older observations; $d = 1$

gives a uniform moving average. The study fixes $M = 252$. It does not infer d from a Hurst estimator.

The normalized operator is a finite-window analogue of a fractional integral. For a smooth signal q, a physical window T and a refining observation step, the corresponding normalized fractional integral is

$$\mathcal{I}_{d,T}q(t) = \frac{d}{T^d}\int_0^T a^{d-1}q(t-a)\,da. \tag{6}$$

This relation motivates the kernel; the empirical model itself is the finite discrete recursion (5). Finite memory and subunit feedback do not establish asymptotic long-range dependence or convergence to a fractional stochastic PDE.

3.2 Correlation graph and implicit diffusion

Estimate a Ledoit–Wolf covariance matrix [10] after standardizing the fitting-window returns. Convert it to a correlation matrix R and set

$$W_{ij} = \max(R_{ij}, 0)^2 \quad (i \neq j), \qquad W_{ii} = 0. \tag{7}$$

Let $D = \mathrm{diag}(W\mathbf{1})$ and rescale the graph Laplacian by its mean degree:

$$L = \frac{D - W}{\max(J^{-1}\,\mathrm{tr}\,D, 10^{-12})}, \qquad A_\tau = (I + \tau L)^{-1}, \quad \tau \geq 0. \tag{8}$$

For the scalar index, $L = 0$ and $A_\tau = 1$; index results contain no evidence about cross-sector diffusion. For sectors, the update $x = A_\tau m$ is the implicit Euler step $x - m = -\tau L x$ for graph heat diffusion. The diffusion parameter controls smoothing on this estimated network, not a causal contagion rate.

No arbitrary one-dimensional sector ordering is required. Under a permutation matrix Π, the same construction transforms L into $\Pi L \Pi^\mathsf{T}$ and A_τ into $\Pi A_\tau \Pi^\mathsf{T}$. Forecasts inherit this permutation equivariance.

3.3 Conditional variance and return generation

For $0 \leq \eta < 1$, define normalized conditional variance h_t and asset-scale variance v_t by

$$h_t = (1-\eta)\mathbf{1} + \eta A_\tau m_t, \qquad v_t = s^2 \odot h_t, \tag{9}$$

where $\odot$ denotes elementwise multiplication. Returns obey

$$r_t = \mu + \mathrm{diag}(\sqrt{v_t})\,\varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0, \quad \mathbb{E}[\varepsilon_{t,j}^2 \mid \mathcal{F}_{t-1}] = 1. \tag{10}$$

Simulations use temporally independent Gaussian innovation vectors with correlation R; Gaussianity is not required by the marginal moment results.

Diffusion acts on standardized variance signals. It preserves their sum, not necessarily the sum of variances in original asset units. This distinction allows sectors to retain different volatility scales.

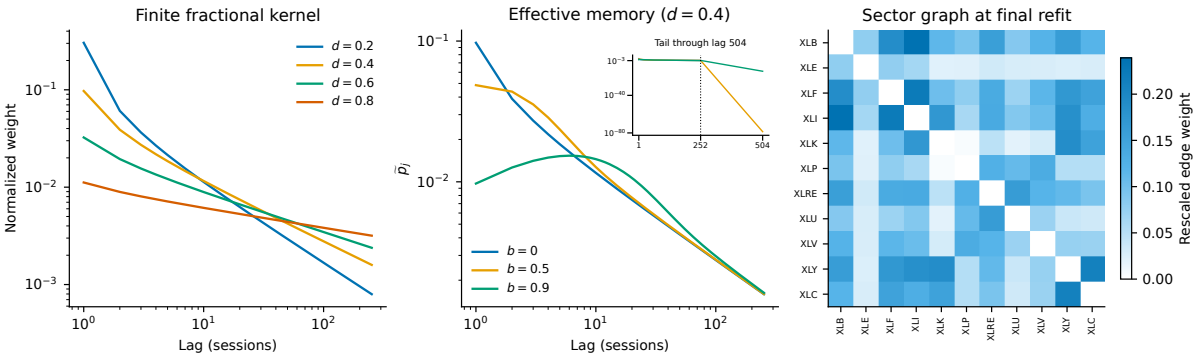


Figure 1: Left: normalized fractional weights over the 252-session window. Center: the effective weights $\widetilde{p}_j$ from (22), illustrated at $d = 0.4$ and $b = 0, 0.5, 0.9$. The main panel shows lags through 252; the inset extends to lag 504. The $b = 0$ curve ends at lag 252; positive persistence adds a geometric tail. These illustrative parameters are not selected fits. Right: the sector graph from the final parameter refit, constructed from rows through that origin.

To make the connection with multivariate ARCH explicit, let $S = \mathrm{diag}(s_1^2, \ldots, s_J^2)$. In asset units, the lag coefficient matrices are $C_k = \eta p_k S A_\tau S^{-1}$, giving

$$v_t = (1-\eta)s^2 + \sum_{k=0}^{M-1} C_k\Big(u_{t-1-k} \odot u_{t-1-k}\Big).$$

All C_k are nonnegative, and $\rho(\sum_k C_k) = \eta$. Fractional weighting and graph diffusion constrain these matrices to an interpretable shared structure.

4 Mathematical properties

Proposition 1 (Positive, nonamplifying graph diffusion)**.** *For a symmetric nonnegative adjacency matrix and $\tau \geq 0$, A_τ is entrywise nonnegative and doubly stochastic. Its eigenvalues lie in $(0, 1]$. For any real vector x,*

$$\min_j x_j \leq (A_\tau x)_i \leq \max_j x_j, \qquad (A_\tau x)^\mathsf{T} L (A_\tau x) \leq x^\mathsf{T} L x. \tag{11}$$

Proof. The Laplacian is symmetric positive semidefinite and $L\mathbf{1} = 0$. Hence $I + \tau L$ is invertible and $A_\tau \mathbf{1} = \mathbf{1}$. For $b \geq 0$, a negative minimum of the solution to $x + \tau L x = b$ would make the corresponding left-hand component negative, a contradiction. Hence $A_\tau \geq 0$. Symmetry gives unit column sums. The componentwise bound follows from convex averaging. Diagonalizing L gives eigenvalues $(1 + \tau\lambda_i)^{-1}$ for A_τ; the energy inequality follows from $\lambda_i/(1 + \tau\lambda_i)^2 \leq \lambda_i$. □

Proposition 2 (Positivity and fixed-parameter moment stability)**.** *Fix $M < \infty$, $s_j^2 > 0$ and $0 \leq \eta < 1$. For nonnegative observed squared innovations, $v_{t,j} \geq (1-\eta)s_j^2$. If the innovation vectors are iid over time with zero means and unit marginal second moments, the normalized squared-return recursion admits a unique causal stationary solution with finite first moments, and its stationary marginal expectations satisfy $\mathbb{E}z_t = \mathbf{1}$. It follows*

that the corresponding stationary returns have variance s_j^2 *and finite second moments* $\mu_j^2 + s_j^2$.

Proof. The lower bound follows from nonnegative weights and nonnegative diffusion. If initial component expectations are bounded by $C \geq 1$, then $\mathbb{E}h_{t,j} \leq (1-\eta) + \eta C \leq C$, and $\mathbb{E}z_{t,j} = \mathbb{E}h_{t,j}$. For two histories driven by the same innovation sequence, put $e_t = \mathbb{E}\|z_t - z'_t\|_1$. Independence of the current innovations from the past, unit marginal second moments and the unit column sums of A_τ give

$$e_t \leq \eta \sum_{k=0}^{M-1} p_k e_{t-1-k}. \tag{12}$$

The largest expected discrepancy contracts by at least η each time an entire block of M initial lags is replaced. Backward iteration from a bounded deterministic history is therefore Cauchy in L^1, yielding a causal stationary limit; the same bound gives uniqueness among finite-first-moment solutions. In stationarity, $b = \mathbb{E}z_t$ solves $b = (1-\eta)\mathbf{1} + \eta A_\tau b$. Since $\rho(\eta A_\tau) < 1$, its unique solution is $b = \mathbf{1}$. □

The moment result concerns the fixed-parameter recursion, rather than the sequence produced by rolling estimation. It supplies finite return second moments; fourth moments and a limit distribution for empirical losses would require additional conditions. Finite-window contraction also distinguishes the fractional shape of the weights from asymptotic long-range dependence.

5 Forecasting and estimation

5.1 Exact multi-step marginal moments

At origin o, let $\bar{z}_{o+\ell|o}$ equal observed z for nonfuture lags, and recursively replace future squared innovations by their conditional expectations:

$$\bar{h}_{o+\ell|o} = (1-\eta)\mathbf{1} + \eta A_\tau \sum_{k=0}^{M-1} p_k \bar{z}_{o+\ell-1-k|o}, \qquad \bar{z}_{o+\ell|o} = \bar{h}_{o+\ell|o}. \tag{13}$$

Linearity and unit innovation second moments make this recursion exact under the fitted variance model. The aggregate forecast is

$$\widehat{V}_{o,h} = s^2 \odot \sum_{\ell=1}^{h} \bar{h}_{o+\ell|o}. \tag{14}$$

With the model's conditional mean, centered increments are martingale differences, so (14) also gives the marginal conditional variance of cumulative future log-returns. Future variances remain random, however, and their sum does not give a Gaussian predictive law for returns. Our forecasts target variance directly.

5.2 Chronological selection

Forecast origins occur every five retained observations on a grid beginning at price position 252; scoring begins in 2023. At the first scored origin and whenever at least 21 observations have elapsed since the last refit, parameters are updated. This gives a 25-observation refit spacing on the five-observation grid, with 38 refits per panel in the reported experiment.

At a refit origin, reserve the final 252 observed returns for validation. Means, scales and graph structure used to compute validation predictions are estimated strictly before that block, using at most 1,008 returns. As validation advances, already observed squared returns may update the memory state. Their future counterparts are never accessed. Select the minimum pooled one-step QLIKE over

$$d \in \{0.2, 0.4, 0.6, 0.8\}, \qquad \eta \in \{0.8, 0.95, 0.99\}, \tag{15}$$

$$\tau \in \{0, 0.25, 1, 4\}. \tag{16}$$

The index fixes $\tau = 0$. Stable candidate order resolves exact ties. After selection, refit means, scales and the graph on the last 1,008 available returns, including validation observations, and forecast subsequent targets. The historical period was available during earlier research iterations, so this chronological evaluation remains a development study.

The selected parameters serve all three horizons, although selection minimizes only one-step loss. That mismatch motivates the multi-horizon refinement in Section 7. All methods are scored at the same origins, excluding end blocks without complete targets for every horizon.

5.3 Benchmarks and ablations

1. **Historical variance:** the current fitting-window sample variance, held constant over each forecast horizon.
2. **EWMA:** squared-innovation averaging with decay selected from $\{0.9, 0.94, 0.97, 0.99\}$ on the same validation block. Its future conditional variance remains constant under its own update equation.
3. **GARCH(1,1):** independent asset models estimated by Gaussian quasi-maximum likelihood [3]. Positivity is imposed and $\alpha + \beta < 0.999$. Two fixed initializations are used; all 456 selected asset-level fits report convergence.
4. **Fractional only:** the proposed kernel and feedback with $\tau = 0$, with its own validation-selected d and η.
5. **Exponential plus graph:** replace the fractional average by EWMA while retaining positive feedback and graph diffusion. Its decay, feedback and diffusion grids have the same candidate count as the fractional graph model.

Each ablation selects its remaining parameters on its own validation losses.

6 Evaluation of the initial specification

6.1 Loss and dependence

The primary score is a normalized QLIKE loss,

$$\mathcal{L}_{o,h,j} = \log\left(\frac{\widehat{V}_{o,h,j}}{hs_j^2}\right) + \frac{Y_{o,h,j}}{\widehat{V}_{o,h,j}}. \tag{17}$$

The normalization is known at the origin and common to all methods. It changes the level, not paired comparisons. QLIKE is appropriate for conditional second-moment forecasting with noisy, conditionally unbiased proxies [16, 17]. Scores can be negative, and ratios of score levels are not reported as percentage improvements. A normalized squared-error diagnostic is also retained in the machine-readable results.

There are 186 common origins from 9 January 2023 to 17 September 2026, with targets through 16 October 2026. The sector panel contributes 2,046 asset–origin pairs per horizon, which must not be treated as independent replications. Twenty-one-observation targets overlap. Paired circular block-bootstrap ranges use 2,000 replicates, seed 1729, and blocks of five or ten origins. Entire sector vectors are resampled together. Each replicate averages the paired loss difference, with negative values favoring the proposed forecast. The 2.5th and 97.5th percentiles give the reported nominal 95% intervals. This resampling preserves dependence within each block and across assets; it does not adjust for model selection, multiple comparisons, nonstationarity or previous researcher exposure. An interval containing zero leaves the direction unresolved under this procedure; it is not an equivalence test. Appendix E adds paired predictive-accuracy tests following Diebold and Mariano [6] and calibration regressions, with dependence-aware covariance estimates. The statistical interpretation in this paragraph applies to all subsequent loss comparisons.

6.2 Pooled comparison

Table 1 shows lower mean loss for fractional diffusion than tuned EWMA at every panel–horizon combination. The one-session differences are -0.18834 for the index and -0.07899 for sectors. GARCH has lower mean QLIKE in all six cases. For sectors, the proposed method's excess losses relative to GARCH are 0.01084, 0.01287 and 0.01752 across the three horizons.

Graph diffusion changes the one-session sector score from 0.6549 to 0.6405. Its gain is much smaller at five observations, and the no-diffusion model is slightly better at 21 observations. Nonzero diffusion is selected at 31 of 38 sector refits, but validation selection is not itself evidence of outer-sample predictive benefit. The exponential graph model outperforms fractional diffusion at five and 21 observations in both panels; the fractional kernel has the lower one-session mean in both panels. Memory shape matters, but no shape dominates all horizons.

With ten-origin blocks, the nominal 95% range for the one-session sector difference from EWMA is $[-0.26077, 0.03470]$. Relative to no diffusion, it is $[-0.03672, 0.00145]$. Both include zero. The difference from GARCH is 0.01084, with range $[-0.03822, 0.04668]$.

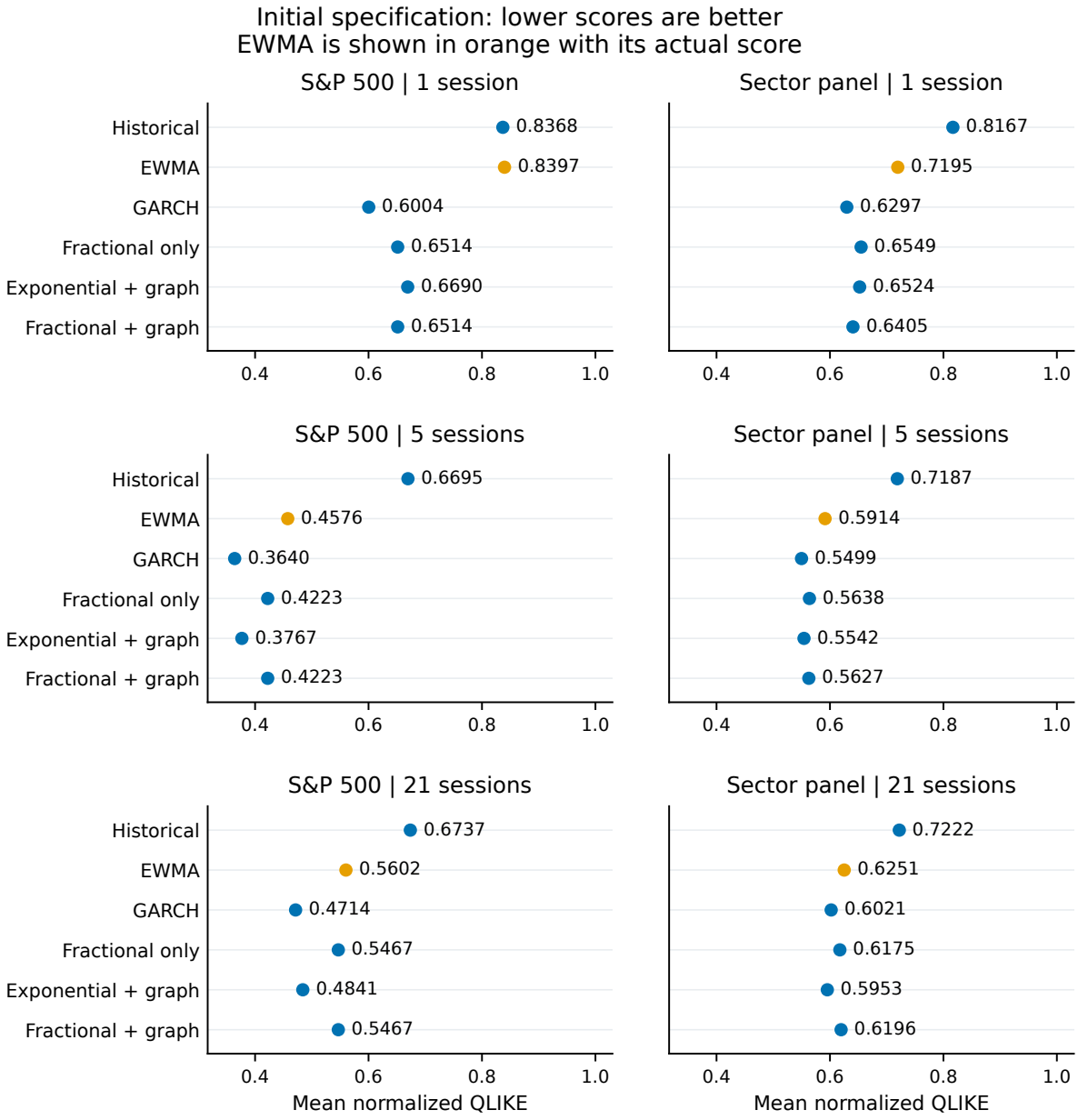


Figure 2: Mean normalized QLIKE for the initial specification and references. Lower values indicate smaller loss. Each point and its numerical label show the actual score, including EWMA in orange. The six panels share the same horizontal scale and reproduce Table 1; all methods use identical origins within each panel and horizon.

	Index			Sectors		
Horizon	1	5	21	1	5	21
Historical	0.8368	0.6695	0.6737	0.8167	0.7187	0.7222
EWMA	0.8397	0.4576	0.5602	0.7195	0.5914	0.6251
GARCH	**0.6004**	**0.3640**	**0.4714**	**0.6297**	**0.5499**	0.6021
Fractional only	0.6514	0.4223	0.5467	0.6549	0.5638	0.6175
Exponential + graph	0.6690	0.3767	0.4841	0.6524	0.5542	**0.5953**
Fractional + graph	0.6514	0.4223	0.5467	0.6405	0.5627	0.6196

Panel	Horizon	Fractional − GARCH	Nominal 95% interval
Index	1	+0.05103	[−0.07687, +0.17028]
Index	5	+0.05823	[−0.00257, +0.12630]
Index	21	+0.07532	[−0.01504, +0.20011]
Sectors	1	+0.01084	[−0.03822, +0.04668]
Sectors	5	+0.01287	[−0.00534, +0.03247]
Sectors	21	+0.01752	[−0.01941, +0.06475]

Table 1: Mean normalized QLIKE at 1, 5 and 21 observations. Lower is better. All methods use identical origins and the same mean convention. The index has no cross-sector diffusion, so its two fractional rows coincide. Bold marks the lowest unrounded loss among the displayed models in each column, including ties. The lower panel adds fractional-plus-graph minus GARCH differences and nominal 95% intervals using ten-origin blocks.

These intervals leave all three rankings unresolved. Full intervals for both block lengths accompany the tables.

6.3 Variation over time

Average improvements are not uniform over calendar years. For the index, the proposed method loses to EWMA at all horizons in 2023, wins at all horizons in 2024 and 2026, and has mixed results in 2025. For sectors, it loses at all horizons in 2023 and improves on EWMA in the later years. The 2025 index one-session difference of −0.5367 contributes substantially to the pooled advantage over EWMA.

Figure 3 shows the corresponding forecast paths. Asset-level losses are retained in the local origin records, and aggregate annual scores are included in the manuscript bundle.

7 Persistence and asymmetric innovations

7.1 Specification and moment properties

The initial model responds identically to positive and negative innovations and selects parameters using only one-step losses. We retain its fractional kernel and graph diffusion

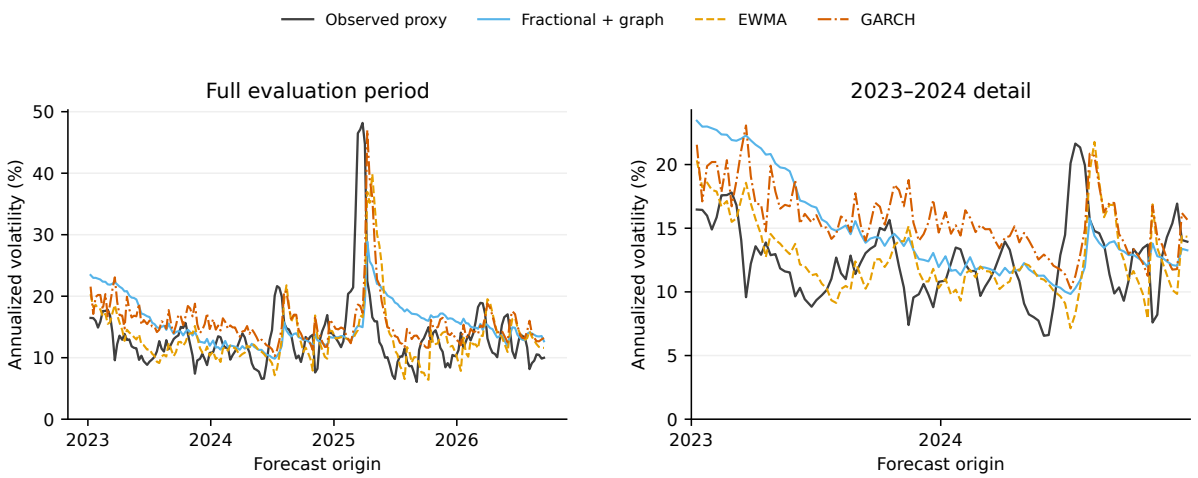


Figure 3: Index risk forecasts over 21 subsequent observations, displayed in annualized volatility units. The black series is the subsequently observed squared-return proxy, plotted at its forecast origin for comparison; it was not available then. The left panel retains the full period; the right enlarges 2023–2024 with its own vertical range. Both use the same saved observations and forecasts.

while allowing a persistent variance state and asymmetric shock contributions. Define

$$x_{t,j} = z_{t,j}\frac{1+\delta\mathbf{1}\{u_{t,j}<0\}}{1+\delta/2}, \tag{18}$$

$$h_t = (1-b)(1-\eta)\mathbf{1} + bh_{t-1} + (1-b)\eta A_\tau \sum_{k=0}^{M-1} p_k x_{t-1-k}, \tag{19}$$

with $v_t = s^2 \odot h_t$, $0 \le b < 1$ and $\delta \ge 0$. The parameter b smooths the variance response. With $\delta = 1$, a negative innovation has twice the contribution of a positive innovation of the same magnitude.

For the asymmetric recursion, assume conditionally symmetric innovations with unit second moments. Conditional symmetry strengthens the moment assumptions: symmetry gives $\mathbb{E}[\varepsilon_{t,j}^2\mathbf{1}\{\varepsilon_{t,j}<0\}] = 1/2$ and therefore $\mathbb{E}[x_t \mid \mathcal{F}_{t-1}] = h_t$. Gaussian simulation innovations satisfy the assumption; it is not asserted as an empirical property of market shocks. Future expected variances are obtained by the same substitution as in Section 5.1, retaining the additional state h_{t-1}.

Unit initial variance and unit presample transformed squares give the floor $v_{t,j} \ge (1-\eta)s_j^2$ by induction. For fixed parameters and iid symmetric innovations, the shared-noise argument for Proposition 2 now gives, for discrepancies between normalized variance processes,

$$e_t \le be_{t-1} + (1-b)\eta \sum_{k=0}^{M-1} p_k e_{t-1-k}, \qquad b + (1-b)\eta < 1. \tag{20}$$

The maximum expected discrepancy over the finite lag state contracts over successive M-observation blocks. The same backward-iteration argument yields a unique causal finite-first-moment stationary variance process. Its mean is $\mathbf{1}$, so return variance is s_j^2. The recursive state adds exponentially decaying persistence; it does not establish asymptotic long-range dependence.

7.2 Effective memory and graph modes

The persistence state has an exact filter interpretation. For fixed parameters, unroll (19) after removing its initialization transient to obtain

$$h_t = (1-\eta)\mathbf{1} + \eta A_\tau \sum_{j=1}^{\infty} \widetilde{p}_j x_{t-j}, \tag{21}$$

$$\widetilde{p}_j = (1-b) \sum_{k=0}^{\min(M-1,j-1)} p_k b^{j-1-k}. \tag{22}$$

Indeed, repeated substitution expands the state into a geometric sum; grouping terms with the same innovation lag yields (22). Since all terms are nonnegative, summing in either order gives $\sum_{j\geq 1} \widetilde{p}_j = (1-b)\sum_k p_k \sum_{\ell\geq 0} b^\ell = 1$. For $0 < b < 1$ and $j \geq M$, $\widetilde{p}_{j+1} = b\widetilde{p}_j$. At $b = 0$ the weights reduce to p_{j-1} for $1 \leq j \leq M$, with zero tail. The refinement geometrically smooths the finite fractional kernel; its infinite tail, when present, is exponential rather than hyperbolic.

With $L = U\Lambda U^\top$, the graph-mode coefficient at lag j is $\eta\widetilde{p}_j/(1+\tau\lambda_\ell)$. Temporal weighting and spatial smoothing are separable: every lag shares the same graph attenuation. This clarifies both interpretability and a restriction relative to more general network lag matrices. A numerical impulse-response test verifies the identity against the implemented filtering recursion.

7.3 Recorded follow-up protocol

The refinement was designed after inspecting the initial study. `paper/refinement_protocol.json` records its rules before candidate evaluation on the reused historical targets. At each refit, use the same 252-observation validation block and fit nuisance parameters strictly before it. Retain only validation origins whose complete 21-observation targets finish by the refit date. Minimize the equal-weight mean of normalized QLIKE at horizons 1, 5 and 21, using all eligible daily origins. A single selected specification serves all outer horizons.

Retain the earlier grids for d, η and τ, and add $b \in \{0, 0.5, 0.9\}$ and $\delta \in \{0, 1\}$. There are 288 candidates for sectors and 72 for the scalar index per kernel family. A symmetric variant fixes $\delta = 0$; a graph ablation fixes $\tau = 0$; a kernel ablation uses normalized exponential weights truncated to the same 252 observations, with the earlier four decay values. Each ablation has its own validation selection. The larger candidate set increases selection risk despite chronological ordering.

An additional GJR-GARCH reference permits asymmetric shock responses [7]. Its normalized variance satisfies $h_t = \omega + (\alpha + \gamma\mathbf{1}\{u_{t-1} < 0\})z_{t-1} + \beta h_{t-1}$. Independent asset fits use Gaussian QMLE, two fixed starts, nonnegative coefficients and $\alpha + \beta + \gamma/2 < 0.999$. All 456 selected fits report convergence. Multi-step moments also require conditional symmetry. Earlier reference forecasts are recomputed; all 40,176 original forecast records and loss values match exactly, so comparisons use unchanged targets and scales.

	Index			Sectors		
Horizon	1	5	21	1	5	21
Fractional + graph	0.6514	0.4223	0.5467	0.6405	0.5627	0.6196
Persistent symmetric	0.6143	0.3815	0.5162	0.6447	0.5582	0.6116
Refined fractional	0.5819	0.3574	0.4977	0.6242	0.5478	0.6043
Asymmetric, no graph	0.5819	0.3574	0.4977	0.6317	0.5474	0.6046
Asymmetric exponential	0.6223	0.3516	0.4785	0.6656	0.5588	0.5980
GARCH	0.6004	0.3640	0.4714	0.6297	0.5499	0.6021
GJR-GARCH	**0.5251**	**0.3128**	**0.4360**	**0.6053**	**0.5401**	**0.5931**

Table 2: Mean normalized QLIKE for the refinement, its ablations and references. "Fractional + graph" denotes the initial variance specification. All rows use the same 186 origins. Lower is better; bold marks each column's minimum.

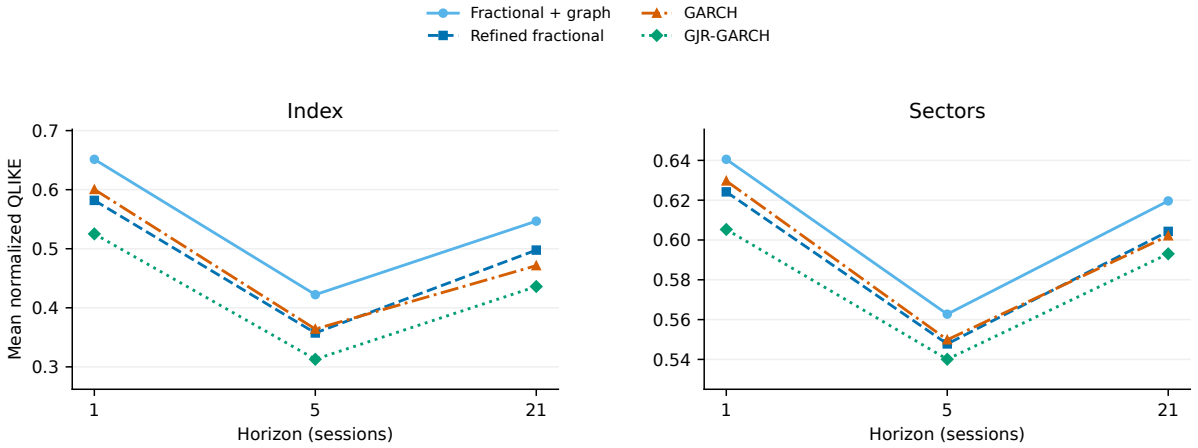


Figure 4: Pooled losses before and after the refinement, alongside GARCH and GJR-GARCH. The asymmetric reference remains stronger in every displayed case.

7.4 Follow-up results

The refinement lowers pooled loss in all six comparisons with the initial model (Table 2). It also improves on GARCH at one and five observations in both panels, but GARCH remains better at 21 observations and GJR-GARCH at every panel–horizon combination. Persistence, asymmetry and multi-horizon selection jointly improve the fractional family.

Using ten-origin blocks, nominal 95% ranges for the change relative to the initial fractional model exclude zero in five of six comparisons; the exception is one-session sector forecasting. For the index, the one-session change is -0.06952 with range $[-0.14531, -0.00876]$. All ranges comparing the refinement with GARCH include zero. Their interpretation follows the exploratory bootstrap convention in Section 6.1.

Asymmetry is selected at all 38 index refits and 35 sector refits; nonzero variance persistence is selected at every refit. The graph is selected with nonzero strength at 18 sector refits. Relative to its graph ablation, the refinement improves the one-session sector mean by 0.00752, slightly worsens the five-session mean and changes the 21-session mean little. The exponential ablation is stronger for index horizons 5 and 21 and sector horizon 21.

The ranking depends on both the component and the horizon. Annual improvements also vary: relative to the first specification, index loss rises at 21 sessions in 2024, and sector loss rises at 5/21 sessions in 2024 and at one session in 2025. All annual results remain in the public tables.

8 Extension to country ETFs and additional benchmarks

8.1 A specification fixed before the new download

The previous two studies reuse US history inspected during development. To assess transfer to another universe, a third protocol fixed eight country ETFs before downloading their prices: EWA (Australia), EWC (Canada), EWG (Germany), EWH (Hong Kong), EWJ (Japan), EWL (Switzerland), EWQ (France) and EWU (United Kingdom). The long-established US-listed iShares equity funds were selected for geographical spread. The request covers 1 January 2018 through 16 October 2026, using the same adjusted-price convention. All eight assets are required; missing prices are not filled and assets are not removed after seeing results.

The unchanged fractional implementation, parameter grids, validation windows, refit schedule and three horizons are frozen in `paper/external_protocol.json`, with source hashes. No external loss is used to revise that specification. The timestamp is a local audit record. The country panel had not been evaluated in this project, but it shares global shocks, US trading hours and dollar denomination with the development assets. This experiment tests transfer to another asset universe on the same historical period; it does not supply independent or prospective confirmation.

An archived amendment records a FIGARCH array-layout correction made before its comparison results, without changing model or evaluation rules.

8.2 Additional forecaster and fractional benchmark

We also evaluate a fixed combination of variance forecasts:

$$\widehat{V}^{\mathrm{ens}}_{o,h,j} = \tfrac{1}{2}\widehat{V}^{\mathrm{frac}}_{o,h,j} + \tfrac{1}{2}\widehat{V}^{\mathrm{GJR}}_{o,h,j}\,. \tag{23}$$

The fractional component is the persistent asymmetric model with selected diffusion. Equal weighting, fixed before the country-ETF download, preserves positivity and avoids estimating combination weights [20]. QLIKE is evaluated on the combined variance. Both components are reported separately to assess whether fractional forecasts add value beyond GJR.

FIGARCH(1,d,1) [1] is added as an estimated benchmark using `arch` 7.2.0, Gaussian QMLE and the library's 1,000-lag ARCH approximation. Each asset is centered and standardized using the same origin-fitted mean and scale as the other methods. Fit on the preceding 1,008 returns at the same refit dates, with default starting estimates and SLSQP limited to 1,000 iterations. Between refits, keep parameters and the fitting backcast fixed

while updating on observed returns. Analytic forecasts are transformed back to asset units and summed over the same horizons. Nonfinite results abort the experiment; finite nonconverged fits are reported rather than silently replaced. FIGARCH's approximation and initialization differ from our normalized 252-lag kernel and are recorded explicitly.

The ensemble and FIGARCH are also evaluated against the recorded US forecasts, after verifying code and input hashes. These exploratory comparisons retain the common targets and bootstrap convention of Section 6.1.

8.3 Results

The international panel contains 2,210 prices per asset. Evaluation has 186 origins from 6 January 2023 to 16 September 2026, with complete targets through 15 October 2026 and 38 parameter refits. Its five-observation grid has a different calendar phase from the US grid; comparisons are paired within panels. All 760 FIGARCH asset fits across the three panels converge, as do all 304 international GARCH and 304 GJR-GARCH fits.

The unchanged fractional model obtains international QLIKE $(0.52898, 0.66359, 0.67147)$ at horizons $(1, 5, 21)$, below both EWMA and FIGARCH at every horizon. However, GARCH and GJR-GARCH are stronger at all three. The no-graph ablation has lower loss throughout, while the exponential kernel is better at horizons 5 and 21. Thus the second universe does not support a general benefit from diffusion or a uniform fractional-kernel advantage.

The fixed ensemble lowers those fractional scores to $(0.51307, 0.65094, 0.66134)$, improving all three external pooled comparisons. Its asset-level loss is lower in 22 of 24 country–horizon comparisons, but annual pooled loss worsens relative to the standalone fractional model at horizons 5 and 21 in 2024. Across the US and international panels, the ensemble improves on both the standalone fractional model and FIGARCH in all nine panel–horizon comparisons. These are loss differences, not percentages of an arbitrarily normalized score.

Relative to GJR-GARCH, the ensemble wins only for US sectors at 5 and 21 observations and for the international panel at 21 observations. The international 21-session difference is -0.00092, with ten-origin-block nominal 95% range $[-0.01174, 0.01110]$. All nine ensemble-versus-GJR ranges include zero. For international one-session forecasts, the change from the standalone fractional model is -0.01590; the interval is $[-0.03100, -0.00219]$.

Compared with FIGARCH, the difference is -0.03073. Its interval runs from -0.05536 to -0.01145. These descriptive intervals support reporting the observed improvement without claiming reliable superiority over GJR-GARCH. Full annual, asset-level and paired comparisons remain in the public outputs.

8.4 Direct comparison with Network GARCH

A follow-up benchmark evaluates the first-order recursion of Zhou et al. [22], equation (2.2), on the same sector and country panels:

$$h_t^N = \omega \mathbf{1} + \alpha z_{t-1} + \lambda B z_{t-1} + \beta h_{t-1}^N, \qquad \omega > 0, \quad \alpha, \lambda, \beta \geq 0, \quad \alpha + \lambda + \beta < 0.999. \tag{24}$$

	Countries		
Horizon	1	5	21
EWMA	0.5321	0.6700	0.6823
GARCH	0.5287	0.6587	0.6672
GJR-GARCH	**0.5083**	**0.6496**	0.6623
FIGARCH	0.5438	0.6671	0.6730
Asymmetric, no graph	0.5234	0.6633	0.6701
Asymmetric exponential	0.5290	0.6526	0.6653
Refined fractional	0.5290	0.6636	0.6715
Fixed 50:50 ensemble	0.5131	0.6509	**0.6613**

Table 3: Mean normalized QLIKE on the eight pre-specified country ETFs. The fractional row is the unchanged persistent asymmetric model. The fixed ensemble averages its variance forecasts with GJR-GARCH. Lower is better; bold marks the minimum in each column.

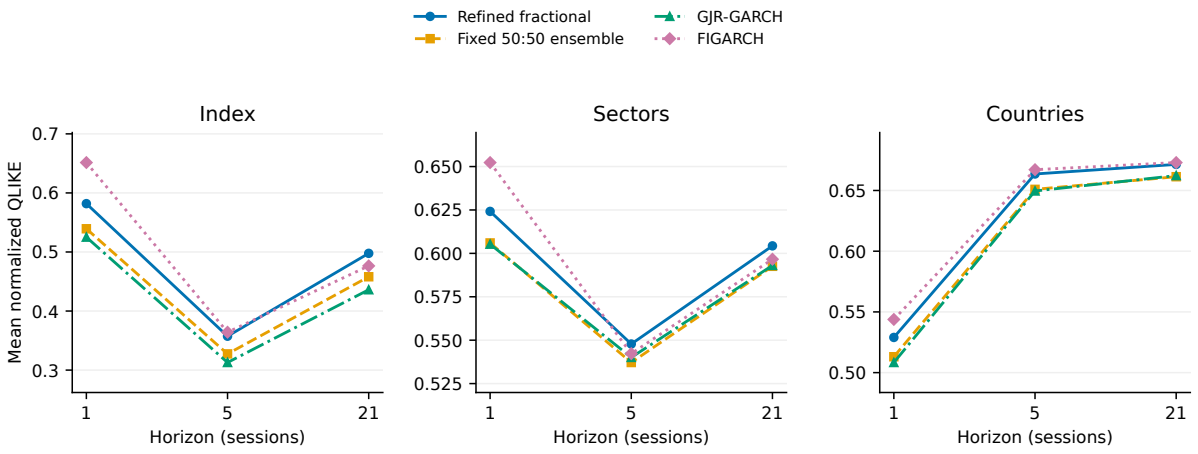


Figure 5: Standalone fractional forecasts, the fixed ensemble, GJR-GARCH and FIGARCH. The index and sector panels reuse the development snapshot; the international panel follows the separately recorded external-universe protocol.

Here B row-normalizes the positive squared correlation weights already used to construct the fractional model's graph. Its diagonal and isolated rows are zero. A sensitivity benchmark replaces these weights by binary edges at $R_{ij} \geq 0.5$ before normalization; a pooled GARCH control sets $\lambda = 0$. The coefficients are shared across the standardized assets. These choices adapt the published recursion to estimated correlation networks rather than the original shareholder network. They do not reproduce the original paper's empirical application or its assumptions of an exogenous network.

At each of the 38 retained refits per panel, marginal Gaussian QMLE uses the preceding 1,008 returns, the common fitted means and scales, and initial normalized variance one. Three fixed starting points are optimized with analytic gradients; the fit with the lowest finite training objective is retained. Observed returns update the state between refits. The first future variance follows (24); subsequent conditional expectations replace future z_t by its conditional mean h_t^N. Daily forecasts are summed over the original horizons and evaluated on identical targets. All 228 selected fits and all 684 optimization starts converge.

The benchmark protocol was recorded before fitting, after earlier results were known. It reuses unchanged forecasts and targets and excludes the scalar index. Annual and asset-level scores and all fitting attempts are retained.

	Sectors			Countries		
Horizon	1	5	21	1	5	21
Refined fractional	0.6242	0.5478	0.6043	0.5290	0.6636	0.6715
Fixed 50:50 ensemble	0.6060	**0.5371**	**0.5925**	0.5131	0.6509	**0.6613**
GJR-GARCH	**0.6053**	0.5401	0.5931	**0.5083**	**0.6496**	0.6623
FIGARCH	0.6523	0.5422	0.5966	0.5438	0.6671	0.6730
Network GARCH (weighted)	0.6149	0.5465	0.6004	0.5397	0.6642	0.6741
Network GARCH (binary)	0.6172	0.5490	0.6029	0.5401	0.6648	0.6750
Pooled GARCH	0.6321	0.5513	0.5997	0.5296	0.6614	0.6725

Table 4: Mean normalized QLIKE for the separately recorded network comparison. The columns retain the same 1-, 5- and 21-session targets. Network GARCH uses shared coefficients; the pooled control removes its network term. Lower is better; bold marks the minimum in each column.

The weighted Network GARCH benchmark has lower pooled loss than the standalone fractional model at all three sector horizons; the ordering reverses at all three country horizons. Every paired 95% range for these six differences includes zero at both block lengths. The fixed ensemble has lower pooled loss than either network variant and the pooled control in all six comparisons. Its one-session country difference against weighted Network GARCH is -0.02660, with ten-origin-block range $[-0.04645, -0.00747]$; the five-session range excludes zero with ten-origin blocks but includes it with five-origin blocks. The other four ranges include zero at both lengths.

Relative to pooled GARCH, weighted network feedback helps the sector mean at one and five sessions, but worsens it at 21 sessions and at all country horizons. All six paired ranges include zero. Binary network losses are higher than weighted network losses in each

comparison. This additional experiment therefore supports a context-dependent role for cross-asset feedback while leaving GJR-GARCH a strong reference.

8.5 Observed evolution and rolling predictions

Figures 6 and 7 compare predictions and subsequent squared-return targets in annualized volatility units:

$$\widehat{\sigma}^{\mathrm{ann}}_{o,h,j} = 100\sqrt{\frac{252}{h}\widehat{V}_{o,h,j}}, \qquad \sigma^{\mathrm{proxy}}_{o,h,j} = 100\sqrt{\frac{252}{h}Y_{o,h,j}}. \tag{25}$$

The square root of expected variance differs from expected volatility. At one session, the observed proxy is a scaled absolute centered return.

The time-series panels use the *end of the target window* on the horizontal axis. A prediction plotted at $o + h$ is computed from rows through o in the recorded snapshot. Curves join the retained origins, spaced five observed sessions apart; they are not a new forecast issued every day. No further fitting, time smoothing, clipping of peaks or favorable subperiod selection is applied. The 21-session target windows overlap.

Multi-asset displays average variances before applying (25). Appendix D retains every ETF's 21-session path to expose errors masked by aggregation.

The index curves also reveal limitations hidden by pooled scores. The abrupt 2025 increase is incompletely anticipated, and the persistent forecasts can remain elevated after the observed proxy falls. Smoother predictions should not be interpreted as pointwise reproduction of the realized fluctuations.

9 Interpretation, limitations and conclusions

9.1 What the restrictions explain

Theoretical contribution. The model restricts multivariate variance lag matrices to a common positive graph resolvent and a shared fractional profile. This structure yields an explicit impulse response and fixed-parameter moment bounds. It excludes asset-specific lag shapes and lag-dependent network geometry. The persistent state smooths the finite fractional profile geometrically, producing an infinite exponential tail when its coefficient is positive.

Empirical contribution. Persistence and asymmetry improve the initial model, while the benefits of diffusion and fractional weighting vary across panels and horizons. Combining the model with GJR lowers loss relative to its standalone forecasts and FIGARCH, but establishes neither superiority nor equivalence to GJR. Table 5 summarizes the comparisons. The calibration diagnostics also reveal one-session country-panel miscalibration for all three examined models, including GJR (Appendix E).

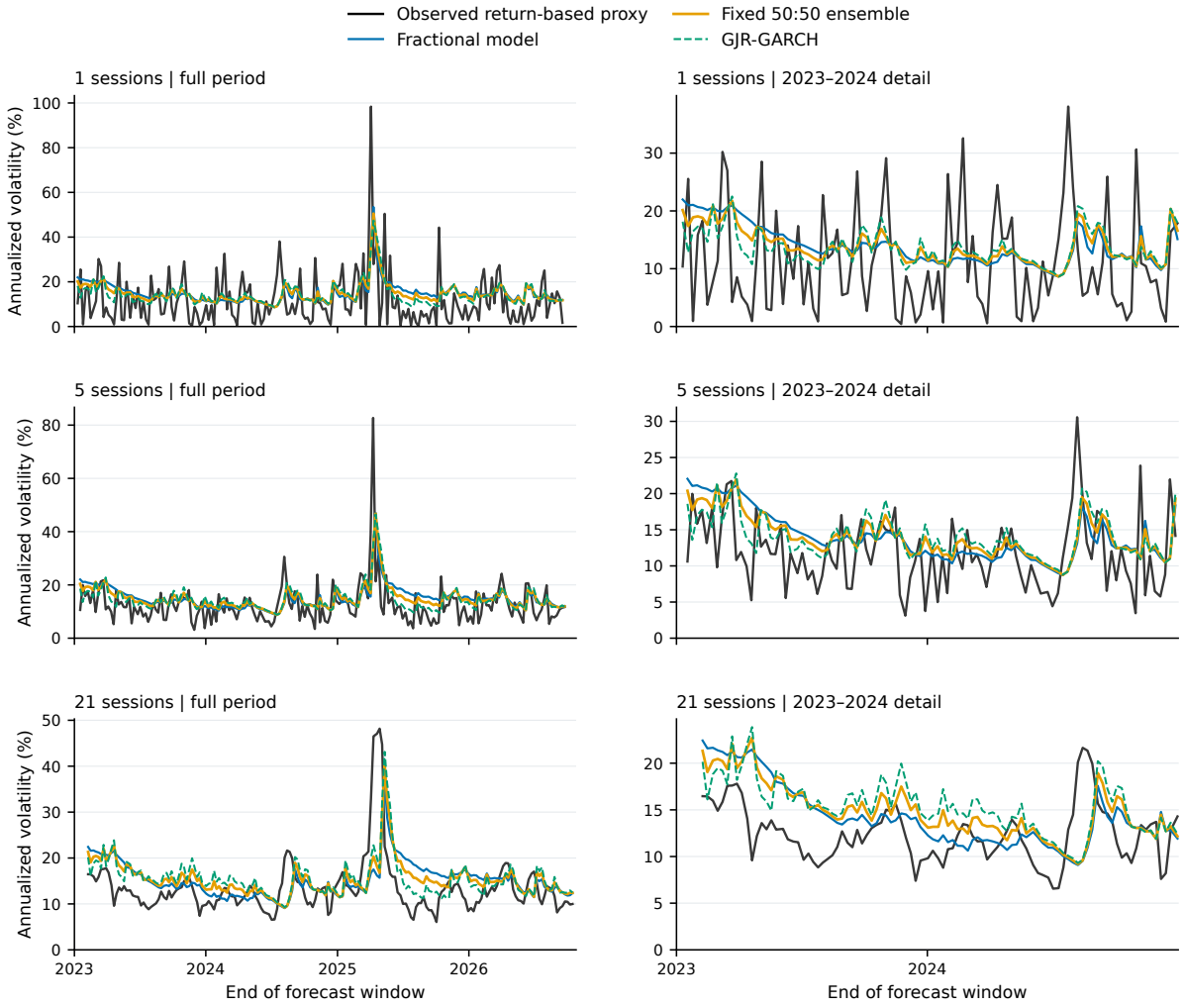


Figure 6: S&P 500 observed proxy and rolling predictions at 1, 5 and 21 sessions. Black denotes the subsequent return-based observation; blue is the persistent fractional model, orange the fixed ensemble, and dashed green GJR-GARCH. Each forecast was issued the stated number of observations before its plotted window-end date. Left panels retain the full period; right panels enlarge 2023–2024. Vertical ranges adapt to each panel, with all displayed peaks retained.

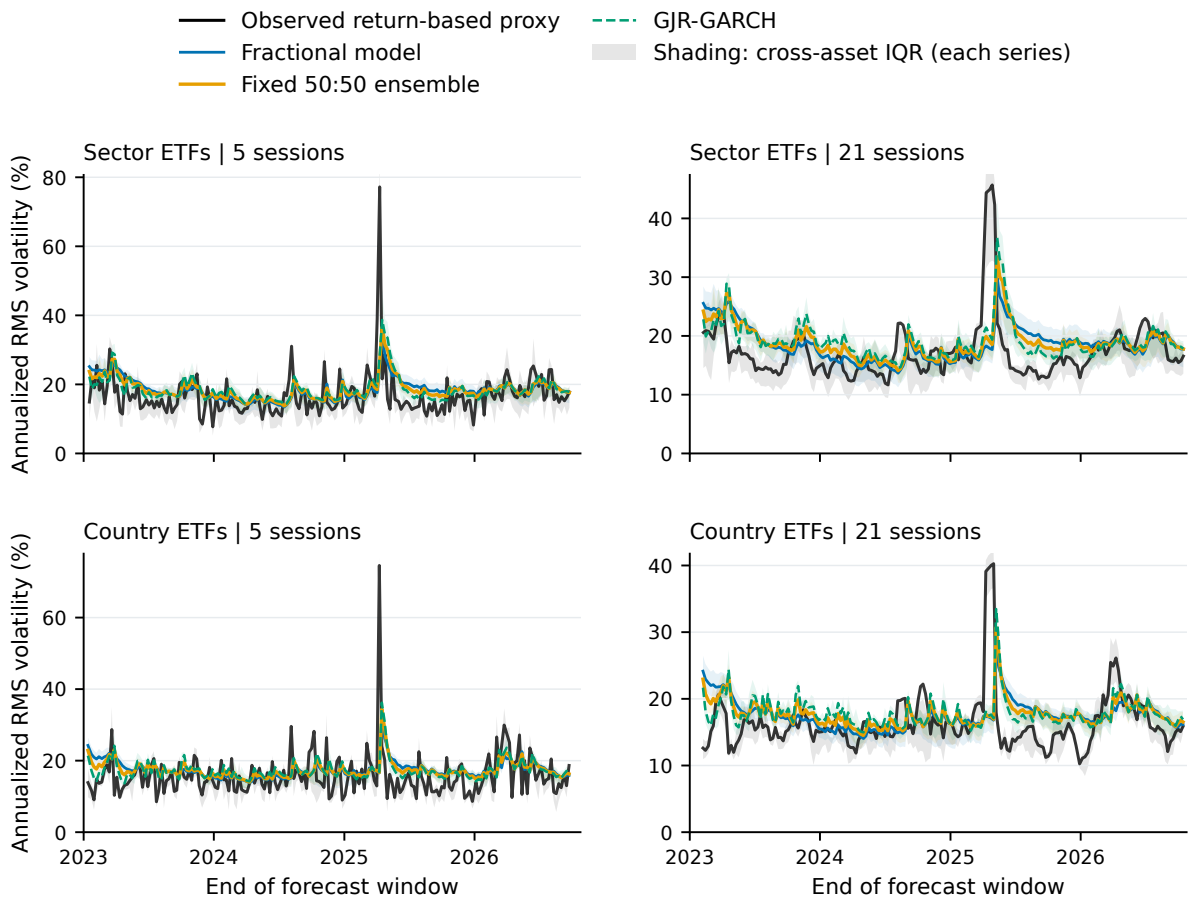


Figure 7: Observed and predicted risk evolution for all eleven sector ETFs and all eight country ETFs, at horizons of 5 and 21 sessions. Marginal variances are averaged across the complete panel before taking the square root. The lines show annualized root-mean-square marginal volatility. Shading in each series' color spans the cross-asset interquartile range of displayed volatilities at each target date; it describes dispersion across ETFs, not forecast uncertainty. Covariances are not included in this display.

Question	Recorded finding	Interpretation
Persistence, asymmetry and horizon selection	Lower mean loss in 6/6 US cases	Joint refinement; development data
Fixed ensemble vs. fractional / FIGARCH	Lower mean loss in 9/9 cases for each	Improvement relative to each reference
Fixed ensemble vs. GJR	Lower mean loss in 3/9 cases	All paired intervals include zero
Diffusion on country ETFs	No-graph variant better at all 3 horizons	No general diffusion benefit
Fractional vs. weighted Network GARCH	Lower loss in countries; higher in sectors	All 6 paired intervals include zero
Country one-session calibration	All 3 models reject the joint calibration null	Post-hoc HAC/Wald diagnostic

Table 5: Summary of the recorded findings. Lower mean losses describe this sample; inferential qualifications follow Section 6.1.

9.2 Data vintages and prospective evaluation

Chronological access to a retrospective snapshot does not recover historical data availability. For prices from two download vintages, define $a_t = P_t^{(T')}/P_t^{(T)} > 0$. Their log-returns satisfy

$$r_t^{(T')} - r_t^{(T)} = \log a_t - \log a_{t-1}. \tag{26}$$

A common multiplicative adjustment cancels; changes in adjacent-date factors can alter returns, fitted parameters and targets. The data interface documents possible dividend, split and price corrections [21], without establishing their occurrence in these panels. With only one retained vintage per panel, we cannot measure this sensitivity.

Appendix G fixes a separate future evaluation with archived data vintages, forecasts issued before their target intervals, range-based volatility measurements and explicit failure rules. Its results are pending.

9.3 Scope and reproducibility

Daily squared-return targets are noisy, the conditional mean is constant within each fit, and the universes share market shocks. The code and records permit checks against the equations given the retained inputs. Prospective observations and richer variance measurements are needed to strengthen the evidence; network log-ARCH, threshold Network GARCH and dynamic-covariance benchmarks would test less restrictive dependence structures.

A Implementation and reproduction

The core specifications are implemented in `fractional_diffusion.py` and `refined_fractional.py`; `external_validation.py` and `network_benchmark.py` supply the additional comparisons. The repository's `paper/README.md` gives execution, download and build commands. Protocols record candidate grids and source hashes; retained outputs include selected parameters, graphs, optimizer statuses and origin-level forecasts.

All 79 tests pass on Python 3.12 under macOS. They cover recursions, graph properties, effective memory, moments, benchmark agreement and inference. Truncated-history and future-price perturbation checks verify forecast alignment. Further tests check the diffusion-risk identity and the frozen forward collector's timing and record integrity.

`render_submission_assets.py` regenerates the eleven historical figures and numerical tables from saved records without refitting; manifests record input, source and output hashes. `run_diffusion_risk.py` generates the additional synthetic figure. Offline Gaussian examples check execution and are not evidence of market forecast accuracy.

B Annual scores

	Index			Sectors		
Year	1	5	21	1	5	21
2023	0.0021	-0.0541	-0.0447	0.2066	0.1643	0.1675
2024	0.8771	0.2769	0.3731	0.3427	0.3209	0.3531
2025	1.2053	0.9798	1.3312	1.1208	0.9419	1.1219
2026	0.4703	0.5115	0.5196	0.9897	0.9254	0.9201

Table 6: Annual mean normalized QLIKE for the fractional graph model. Levels reflect both realized fluctuations and the origin's reference scale. Negative scores are permissible; comparisons should use common-origin loss differences.

C Relation to the original model

The original project used signed Grünwald–Letnikov coefficients in a return recurrence and a reflected nearest-neighbor sector Laplacian. Isolating the current value in a GL derivative does not yield that recurrence's lag and sign convention. A fractional derivative also does not inherit the ordinary logarithmic chain rule. The present work therefore starts from (4)–(10), rather than invoking a fractional Black–Scholes equation.

The retained idea is memory plus diffusion. The changes are explicit: positive fractional-integral weights act on squared standardized innovations; a symmetric estimated graph replaces the artificial sector lattice; positive feedback controls second moments; and risk forecasts are compared chronologically against strong variance references. Hurst identification, convergence of a pricing PDE, and directional price-prediction superiority are not claimed.

D Observed and predicted paths for every ETF

The figures retain every ETF over the full period at the 21-session horizon. Dates mark target-window ends; each asset uses its own logarithmic scale in the units of (25), without clipping or extra smoothing.

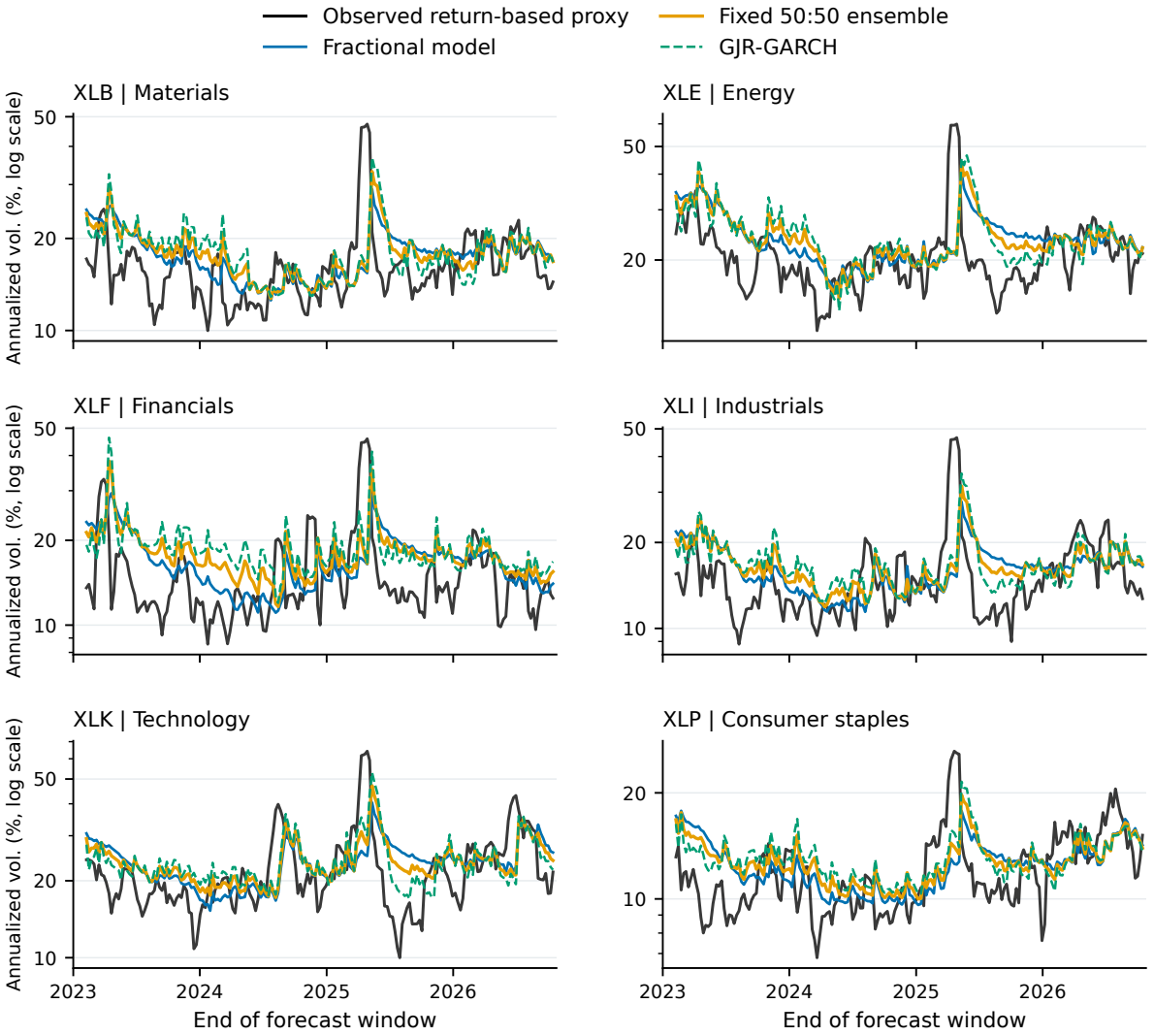


Figure 8: Twenty-one-session observed proxies and forecasts for materials, energy, financials, industrials, technology and consumer staples. Curves use the same methods and colors as Figure 6.

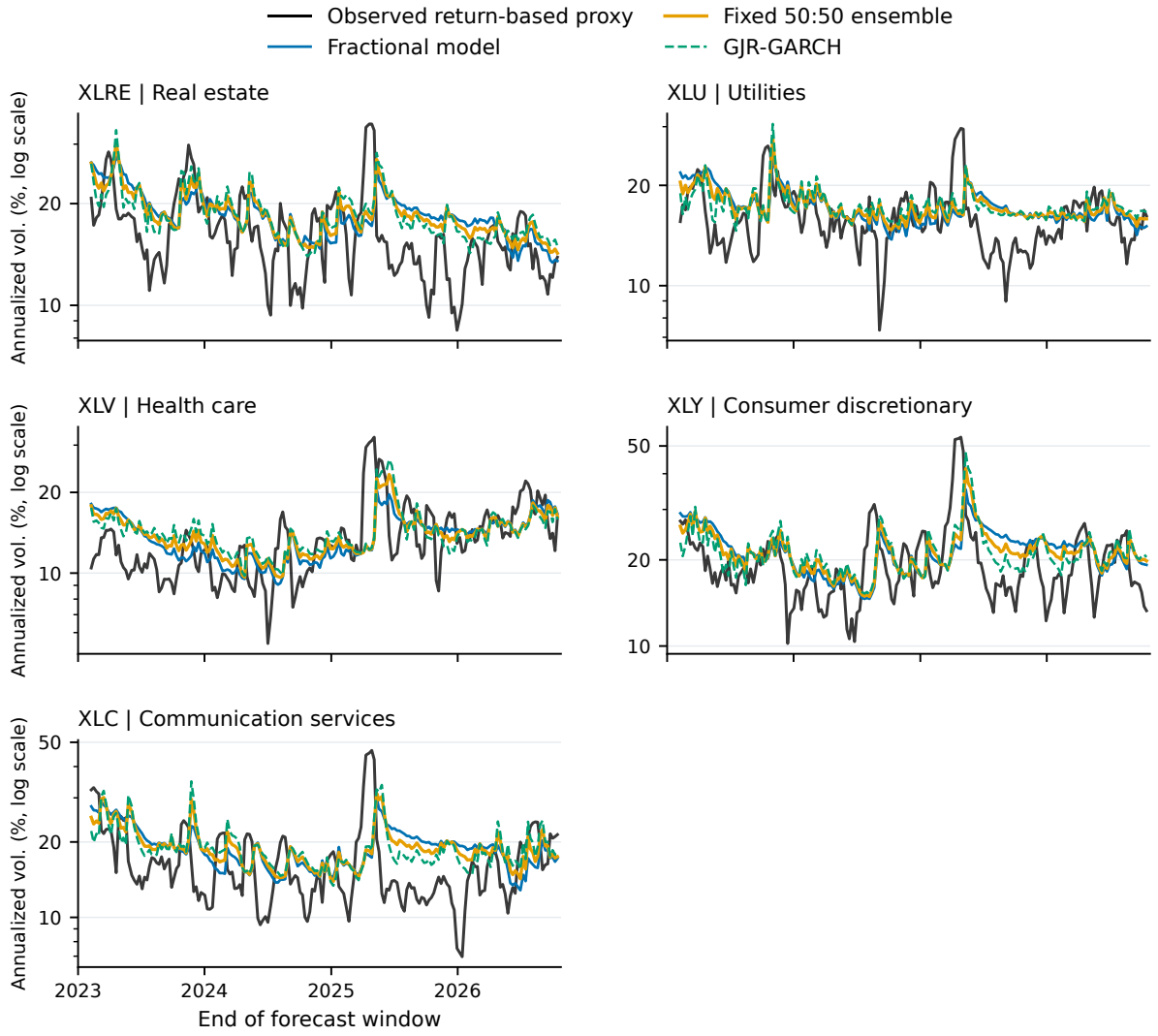


Figure 9: The remaining five sector ETFs: real estate, utilities, health care, consumer discretionary and communication services. Every sector is included exactly once across these two figures.

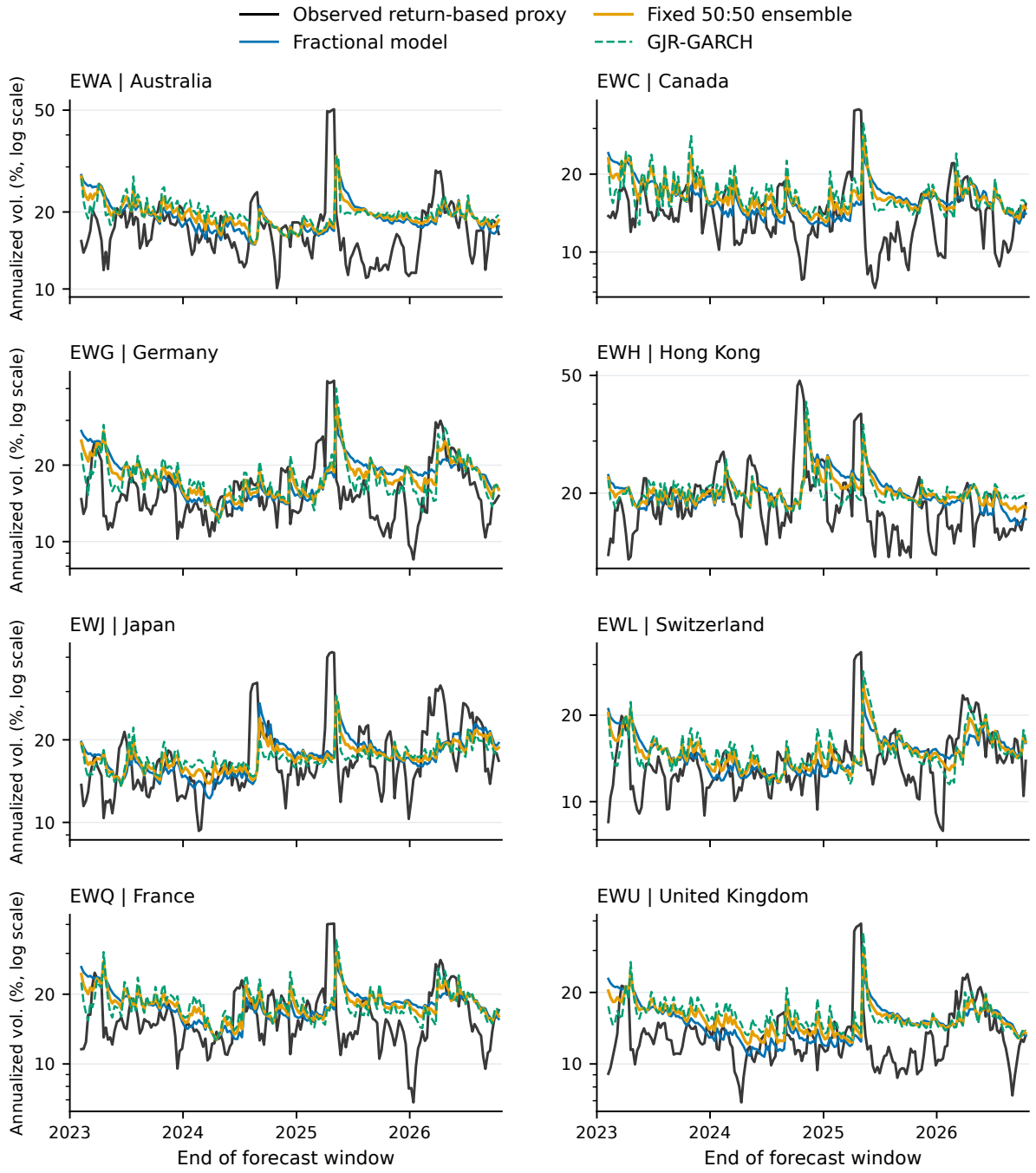


Figure 10: Twenty-one-session observed proxies and forecasts for all eight pre-specified country ETFs. The observed series is a noisy daily-return-based proxy; the plots do not imply prediction of the underlying ETF price direction.

E Additional forecast diagnostics

The diagnostics use saved forecasts and losses without recalibration. Their protocol was fixed after earlier results were known, so they remain exploratory. An archived pre-calculation amendment changed the CSV reconstruction tolerance from 10^{-12} to 10^{-10}; stored losses were unchanged.

E.1 Paired predictive accuracy

For each panel and horizon, let d_o be the asset-average QLIKE difference between a candidate and GJR at origin o. Following Diebold and Mariano [6], we test a zero mean difference using

$$\mathrm{DM}_L = \frac{\bar{d}}{\sqrt{\widehat{\Omega}_L/n}}, \qquad \widehat{\Omega}_L = \widehat{\gamma}_0 + 2\sum_{\ell=1}^{L}\left(1-\frac{\ell}{L+1}\right)\widehat{\gamma}_\ell, \quad \widehat{\gamma}_\ell = \frac{1}{n}\sum_{o=\ell+1}^{n}(d_o-\bar{d})(d_{o-\ell}-\bar{d}).$$

The Bartlett HAC estimator follows Newey and West [13]. We use $n = 186$ origins and two-sided asymptotic normal p-values, with $L = 4$ as the primary choice and $L = 9$ as sensitivity. Lags count forecast origins, spaced five retained sessions apart: a 21-session target overlaps the next four origins. Asset losses are averaged before inference, preserving their contemporaneous dependence within each origin.

Table 7 reports raw and Holm-adjusted [8] values for 18 comparisons; adjustment is separate for each lag choice. None rejects at 5%, even before adjustment. This asymptotic inference requires loss-moment and dependence conditions beyond Proposition 2; the development-data qualification in Section 6.1 also applies.

E.2 Calibration

For method m, average the normalized target and forecast over the J assets at each origin:

$$y_o = \frac{1}{J}\sum_{j=1}^{J}\frac{Y_{o,h,j}}{hs^2_{o,j}}, \qquad x_o^{(m)} = \frac{1}{J}\sum_{j=1}^{J}\frac{\widehat{V}^{(m)}_{o,h,j}}{hs^2_{o,j}}, \qquad y_o = \alpha + \beta x_o^{(m)} + e_o.$$

The Mincer–Zarnowitz regression [12, 17] diagnoses calibration under $H_0 : (\alpha, \beta) = (0, 1)$. Normalization uses the stored origin-specific reference scale. OLS coefficients are reported with Bartlett HAC standard errors, an $n/(n-2)$ covariance multiplier and a joint asymptotic χ^2_2 Wald test. The two lag choices are the same as for DM. Holm adjustment covers the 27 regressions separately at each lag choice.

The one-session country regressions reject the joint null for the fractional model, ensemble and GJR at both lag choices after adjustment. Other regressions do not reject at the adjusted 5% level. These are diagnostics of panel-average normalized second moments; aggregation can mask asset-level miscalibration. The fitted regression is not used to change any forecast.

Panel	h	Model	Δ	DM	p_4	p_9	Holm p_4
Index	1	Fractional	+0.05675	0.866	0.386	0.361	1.000
Index	1	Ensemble	+0.01419	0.329	0.742	0.729	1.000
Index	5	Fractional	+0.04454	1.837	0.066	0.074	1.000
Index	5	Ensemble	+0.01458	1.080	0.280	0.260	1.000
Index	21	Fractional	+0.06166	1.435	0.151	0.190	1.000
Index	21	Ensemble	+0.02200	1.174	0.240	0.297	1.000
Sectors	1	Fractional	+0.01889	1.425	0.154	0.144	1.000
Sectors	1	Ensemble	+0.00072	0.101	0.920	0.917	1.000
Sectors	5	Fractional	+0.00772	0.748	0.455	0.458	1.000
Sectors	5	Ensemble	-0.00293	-0.652	0.514	0.523	1.000
Sectors	21	Fractional	+0.01127	0.620	0.536	0.564	1.000
Sectors	21	Ensemble	-0.00050	-0.064	0.949	0.953	1.000
Countries	1	Fractional	+0.02065	1.392	0.164	0.130	1.000
Countries	1	Ensemble	+0.00474	0.622	0.534	0.487	1.000
Countries	5	Fractional	+0.01401	1.097	0.273	0.299	1.000
Countries	5	Ensemble	+0.00136	0.208	0.835	0.844	1.000
Countries	21	Fractional	+0.00921	0.826	0.409	0.454	1.000
Countries	21	Ensemble	-0.00092	-0.176	0.860	0.874	1.000

Table 7: Post-hoc predictive-accuracy diagnostics against GJR. Negative Δ favors the named model. DM uses $L = 4$; p_4 and p_9 use the indicated HAC lag lengths. Holm p_4 adjusts all 18 primary comparisons. Values are rounded only for display.

E.3 Win counts

Table 9 counts origins at which each asset's candidate forecast has lower QLIKE than GJR. Figure 11 answers a different question: how many assets have a lower full-period mean loss? Neither count is an independent-binomial sample. The CSV retains wins, ties and losses with a 10^{-12} tie tolerance.

Model	h	$\widehat{\alpha}$ (SE)	$\widehat{\beta}$ (SE)	R^2	p_4	p_9	Holm p_4
Index							
Fractional	1	-0.173 (0.334)	1.414 (0.549)	0.197	0.644	0.535	1.000
Ensemble	1	-0.212 (0.325)	1.499 (0.562)	0.204	0.551	0.414	1.000
GJR	1	-0.174 (0.298)	1.475 (0.552)	0.196	0.505	0.394	1.000
Fractional	5	-0.168 (0.270)	1.182 (0.474)	0.187	0.640	0.582	1.000
Ensemble	5	-0.150 (0.254)	1.165 (0.459)	0.202	0.610	0.548	1.000
GJR	5	-0.081 (0.220)	1.076 (0.423)	0.204	0.741	0.704	1.000
Fractional	21	0.174 (0.123)	0.709 (0.268)	0.088	0.353	0.242	1.000
Ensemble	21	0.125 (0.176)	0.761 (0.325)	0.136	0.750	0.595	1.000
GJR	21	0.156 (0.175)	0.702 (0.316)	0.164	0.627	0.416	1.000
Sectors							
Fractional	1	-0.590 (0.438)	1.937 (0.751)	0.092	0.354	0.188	1.000
Ensemble	1	-0.565 (0.444)	1.876 (0.739)	0.109	0.390	0.177	1.000
GJR	1	-0.441 (0.422)	1.684 (0.699)	0.115	0.533	0.286	1.000
Fractional	5	-0.166 (0.232)	1.210 (0.409)	0.104	0.435	0.351	1.000
Ensemble	5	-0.151 (0.247)	1.170 (0.412)	0.128	0.439	0.370	1.000
GJR	5	-0.066 (0.238)	1.039 (0.390)	0.138	0.565	0.558	1.000
Fractional	21	0.164 (0.104)	0.756 (0.200)	0.085	0.288	0.369	1.000
Ensemble	21	0.100 (0.159)	0.818 (0.267)	0.124	0.794	0.738	1.000
GJR	21	0.137 (0.184)	0.747 (0.289)	0.142	0.668	0.554	1.000
Countries							
Fractional	1	0.314 (0.090)	0.441 (0.112)	0.041	< 0.001	< 0.001	< 0.001
Ensemble	1	0.277 (0.093)	0.489 (0.115)	0.059	< 0.001	< 0.001	< 0.001
GJR	1	0.280 (0.101)	0.485 (0.126)	0.071	< 0.001	< 0.001	< 0.001
Fractional	5	0.291 (0.161)	0.607 (0.167)	0.020	0.061	0.085	1.000
Ensemble	5	0.296 (0.178)	0.600 (0.177)	0.024	0.062	0.061	1.000
GJR	5	0.340 (0.171)	0.542 (0.162)	0.025	0.012	0.008	0.287
Fractional	21	0.344 (0.169)	0.527 (0.228)	0.038	0.105	0.135	1.000
Ensemble	21	0.323 (0.213)	0.545 (0.263)	0.046	0.219	0.164	1.000
GJR	21	0.366 (0.218)	0.482 (0.255)	0.047	0.111	0.050	1.000

Table 8: Calibration of normalized panel-average forecasts. Parentheses contain HAC standard errors with $L = 4$. The joint null is $(\alpha, \beta) = (0, 1)$; p_4 and p_9 are raw Wald values, and Holm p_4 adjusts all 27 primary regressions. Small p-values indicate a departure from calibration, not a better forecasting model.

	Fractional wins			Ensemble wins		
Asset	1	5	21	1	5	21
Index						
SP500	79	84	92	80	86	98
Sectors						
XLB	83	108	111	86	110	114
XLE	85	89	99	89	93	102
XLF	91	100	109	95	109	116
XLI	87	97	101	91	100	106
XLK	84	86	95	90	90	103
XLP	106	100	98	107	106	102
XLRE	96	85	82	96	89	87
XLU	97	102	91	98	104	98
XLV	97	111	109	100	118	118
XLY	89	98	86	91	103	88
XLC	80	86	80	84	93	83
Countries						
EWA	85	91	100	86	96	101
EWC	89	85	96	96	95	101
EWG	73	85	85	73	90	89
EWH	98	95	89	99	97	94
EWJ	83	92	94	85	97	102
EWL	88	89	99	90	94	107
EWQ	85	89	99	88	91	107
EWU	87	79	95	90	83	98

Table 9: Per-asset counts of lower-loss origins against GJR, out of 186 paired origins at each horizon. Fractional denotes the refined model. Counts describe frequency; they do not measure the size of gains or losses.

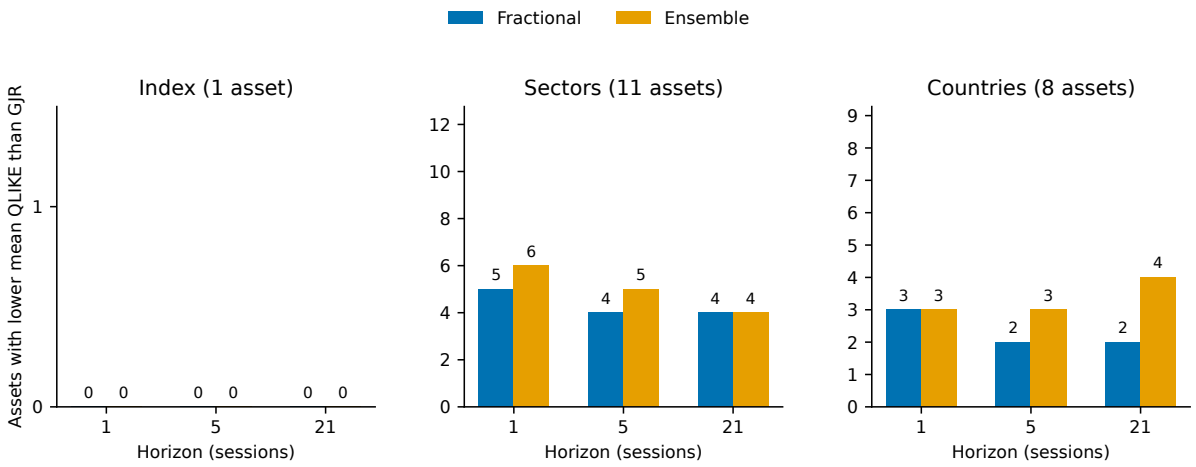


Figure 11: Number of assets with lower full-period mean QLIKE than GJR, by panel and horizon. Denominators are one index, eleven sectors and eight country ETFs. The complete universe is retained in each comparison.

F When diffusion reduces estimation error

The graph can suppress measurement noise while also erasing genuine differences between assets. The following calculation separates these effects. It concerns squared-error estimation of a latent signal with a fixed graph; it is not a claim that the fitted forecasting procedure improves QLIKE. Finite measurement-noise covariance is an additional assumption: second moments of returns alone do not ensure finite variance of squared-return measurements.

Condition on information $\mathcal{G}$ under which a symmetric graph Laplacian L, a signal $m \in \mathbb{R}^J$ and a finite noise covariance Σ are fixed. Suppose $x = m + \varepsilon$, with $\mathbb{E}[\varepsilon \mid \mathcal{G}] = 0$ and $\mathbb{E}[\varepsilon\varepsilon^\top \mid \mathcal{G}] = \Sigma$. Write $A_\tau = (I + \tau L)^{-1}$ and $R_\tau = \mathbb{E}[\|A_\tau x - m\|_2^2 \mid \mathcal{G}]$.

Proposition 3 (Bias and noise reduction under a fixed graph)**.** *For $\tau \geq 0$,*

$$R_\tau = \|(A_\tau - I)m\|_2^2 + \operatorname{tr}(A_\tau \Sigma A_\tau), \tag{27}$$

$$R_\tau - R_0 = \|(A_\tau - I)m\|_2^2 - \operatorname{tr}\{(I - A_\tau^2)\Sigma\}. \tag{28}$$

Consequently, diffusion improves total squared-error risk precisely when its squared bias is smaller than its reduction of noise variance.

Proof. Expand $A_\tau x - m = (A_\tau - I)m + A_\tau \varepsilon$. The conditional cross term vanishes. The noise term is $\operatorname{tr}(A_\tau \Sigma A_\tau)$ by symmetry of A_τ. Subtract $R_0 = \operatorname{tr}(\Sigma)$ and use cyclicity of trace. The spectrum of A_τ lies in $(0, 1]$, so the noise reduction is nonnegative. □

Let $Lv_k = \lambda_k v_k$, with orthonormal v_k, and define $\theta_k = v_k^\top m$, $\nu_k = v_k^\top \Sigma v_k$ and $a_k = (1 + \tau\lambda_k)^{-1}$. The same difference is

$$R_\tau - R_0 = \sum_{k=1}^{J} \left\{ (1 - a_k)^2 \theta_k^2 - (1 - a_k^2)\nu_k \right\}. \tag{29}$$

This identity does not require L and Σ to commute. Modes with $\lambda_k = 0$ are unchanged. For $\tau\lambda_k > 0$, smoothing improves the individual mode precisely when

$$\theta_k^2 < \left(1 + \frac{2}{\tau\lambda_k}\right)\nu_k.$$

The sum in (29), rather than improvement of every mode, is the necessary and sufficient aggregate condition. At zero strength, $R_0' = -2\operatorname{tr}(L\Sigma)$: sufficiently small positive diffusion helps if this trace is positive. At large strength, the estimator approaches projection onto the graph nullspace and may retain substantial bias.

For a fixed intercept and previous state, the one-step forcing $c\mathbf{1} + bh + aA_\tau x$ estimates $c\mathbf{1} + bh + am$ with risk $a^2 R_\tau$. The statement does not compare recursively refitted processes with different past states. Nor does it automatically apply when L is estimated from the same noisy observations: conditioning on that graph need not preserve zero-mean noise. A graph estimated on an earlier block or an explicitly specified conditional noise model would be needed for that interpretation. If the target itself contains noise correlated with ε, the forecast-risk comparison has an additional covariance term. These restrictions separate the mechanism calculation from an empirical performance guarantee.

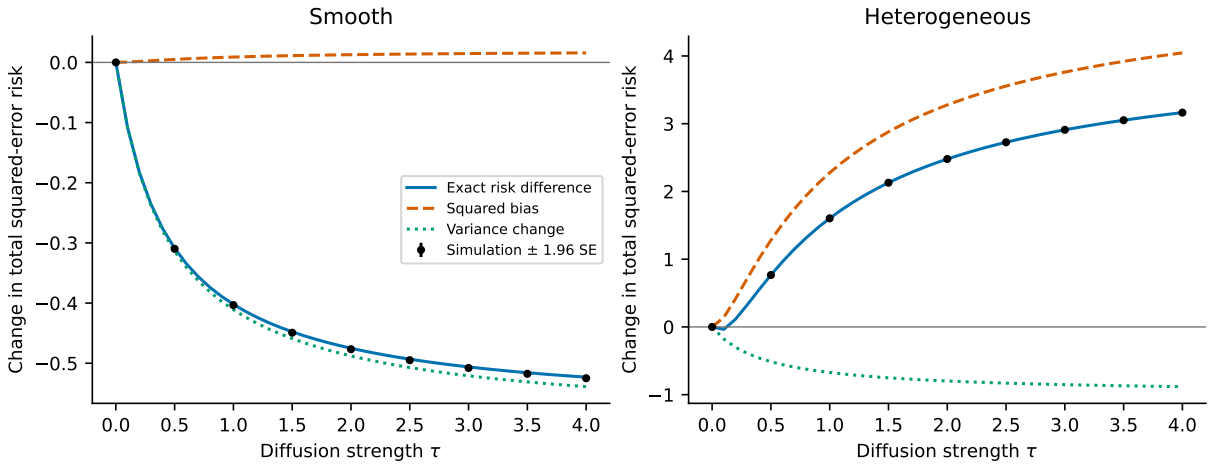


Figure 12: Bias and noise reduction under fixed graph smoothing. Negative total risk differences favor diffusion. Black markers show Monte Carlo differences and 1.96-standard-error bars; the other curves are analytic. Strong smoothing helps similar signals but harms sufficiently heterogeneous ones.

Figure 12 checks the identity in a positive-signal experiment. An eight-node cycle has edge weights $1/2$, with $m_j = 1 + q(-1)^j$ and independent measurements $x_j = m_j \chi^2_{24}/24$. Hence $\Sigma = \mathrm{diag}(2m_j^2/24)$. The two fixed amplitudes $q = 0.05$ and $q = 0.8$ represent similar and heterogeneous signals. At $\tau = 1$, exact total risk changes from 0.668333 to 0.266239 in the first case and from 1.093333 to 2.696557 in the second. The calculation uses 40,000 draws with seed 20261026 and common random numbers across strengths. All plotted risk differences agree with their analytic values within six Monte Carlo standard errors. These are synthetic estimation results, not new market forecast scores.

G Frozen forward evaluation

A separate protocol fixes future evaluation rules before its first target window. It retains the model and selection grids in the main study and adds an OHLC-based measurement of volatility. No completed forward evaluation is reported here. The public protocol, code hashes and calendar are supplied as `prospective_protocol.json`; collection records are kept separately from all historical experiments.

Forecasts and comparisons. The scheduled design uses 52 origins five exchange sessions apart, beginning on 27 October 2026, and 25-session refits. One panel retains the eleven sector ETFs; a second comprises AGG, IEF, TLT, TIP, GLD, SLV, DBC and VNQ, extending the asset mix to bonds, metals, commodities and real estate. Universe membership reflects asset classes rather than measured forecasting gains. These panels remain exposed to common market shocks.

The fractional model, kernel grids, 252-observation validation, 1008-observation estimation window and mean/scale conventions are unchanged. Freezing refers to the specification and update rules; fitted coefficients may change at the scheduled refits. In addition to the selected model and conventional references, matched counterfactuals set only diffusion to 0, 0.25, 1 or 4, retaining the selected kernel, feedback, persistence, asymmetry, mean and scale. This isolates the effect of diffusion from reselection of other parameters.

An evening forecast must be archived by 23:59:59 New York time at the origin. Because an overnight return starts at the preceding close, the design excludes the immediately following session from the target: an h-session forecast is $\sum_{k=2}^{h+1} \widehat{v}_{o+k|o}$ for $h \in \{1, 5, 21\}$. The first target interval therefore starts at the next session's close, after issuance. This one-session embargo is explicit; these scores must not be pooled with the historical study's immediate-next-session targets.

Volatility measurements. For consistently adjusted daily open, high, low and close prices (O_t, H_t, L_t, C_t), define

$$g_t = \log(O_t/C_{t-1}), \tag{30}$$

$$\mathrm{RS}_t = \log(H_t/O_t)\log(H_t/C_t) + \log(L_t/O_t)\log(L_t/C_t), \tag{31}$$

$$Y^{\mathrm{range}}_{o,h,j} = \sum_{k=2}^{h+1} \left\{ (g_{o+k,j} - \widehat{\mu}^{\mathrm{gap}}_{o,j})^2 + \mathrm{RS}_{o+k,j} \right\}. \tag{32}$$

The gap mean is estimated on the common trailing estimation window at refit and then held fixed. The Rogers–Satchell term estimates intraday variance under a log-Brownian model with drift [18]; observed discrete extrema and departures from that model can introduce measurement error. The sum with the centered overnight square estimates whole-day variance only with suitable conditional centering and no material overnight/intraday covariance. It is not high-frequency realized variance and is not asserted to be a universally unbiased proxy for the existing model. The centered close-to-close squared-return target remains a prespecified sensitivity measure, using exactly the same forecasts and dates. Neither OHLC target is used to tune the model on the forward evaluation.

Data capture and inference. The first complete same-session snapshot received between 17:00 and 23:59:59 New York time is retained with raw prices, adjustment factors recoverable from raw and adjusted closes, request/reception times and hashes. Later corrections do not replace the first capture. Missing sessions never shorten a horizon; failed fits, late forecasts and absent snapshots remain in the coverage record. A missed scheduled refit invalidates its dependent origins until the next scheduled refit. No automatic substitute model is used.

Two primary contrasts are fixed in each panel: matched $\tau = 1$ versus $\tau = 0$ on the five-session range target, and selected fractional versus GJR on the 21-session range target. The four primary tests use two-sided DM statistics with Bartlett HAC lag four and Holm adjustment at 5%; lag nine is a fixed sensitivity. Formal primary tests require all 52 scheduled matched origins, preventing compressed time gaps from entering the HAC calculation. Incomplete coverage produces descriptive scores and an inconclusive primary test. Complete coverage does not itself establish adequate power. Other strengths, horizons and the close-based target are secondary analyses. Reporting follows the fixed last target, without interim performance rankings or outcome-dependent stopping. Local timestamps and hashes support an audit but do not by themselves constitute independent preregistration or external proof of forecast publication time.

Data and code availability

Code, tests, protocols and aggregate results are available at https://github.com/enia-ardid/mathematical_finance. The code and manuscript should be identified by their Git commit. Raw prices and detailed origin-level records are retained locally with source metadata and hashes and are not bundled. Exact numerical reproduction requires those snapshots; a fresh adjusted-price download may differ.

Declaration of competing interests

We declare no competing financial interests or personal relationships that could have influenced this work.

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